SEAT NO. DT-23

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

THIRD YEAR(Data Science)

FALL SEMESTER EXAMINATIONS 2025

Batch 2023

Time : 3 Hours

Dated : 03-DEC-25

Max Marks : 60

Artificial Intelligence & Expert Systems - CT-361

Question No.1 [CLO-1] **[Marks: 12]**

Define uncertainty and describe how expert systems handle it. Consider a scenario in which a patient arrives at a hospital with a fever, and an AI-based diagnostic system is used to determine whether the patient has a viral infection. The system uses the following probabilities:

- P(virus) = 0.12
- P(fever | virus) = 0.85
- P(fever | no virus) = 0.20

a. Using Bayes' Theorem, **find** the probability that the patient has a viral infection given that they have a fever.

b. Explain how the system's decision might change if the fever measurement becomes noisy or unreliable.

Question No.2 [CLO-1] **[Marks: 12]**

What is an Expert System? Explain the complete architecture of an Expert System. By using given scenario for an Expert system, answer the questions below:

"An insurance company uses an Expert System to assess car accident claims. The system evaluates information related to driver fault, weather conditions, speed data, past claim history, and damage severity."

a. Describe the Knowledge Base and Inference Engine for this expert system. **What** type of rules would be stored in the Knowledge Base? Justify your answer.

b. Explain **how** forward chaining and backward chaining can be used in this scenario. **Choose** the technique that would be more suitable for this system and justify your choice.

c. Select a small sample rule set (4 rules) of your own and demonstrate **how** inference would proceed for a claim in which the weather was foggy and the driver had no prior violations.

Question No.3 [CLO-2] **[Marks: 12]**

A delivery robot must navigate a warehouse to reach the goal location H, starting from A. The warehouse is represented as a graph where each edge has an associated travel cost. The robot also has access to a heuristic table that estimates the straight-line distance from each node to H.

- **Graph Edges with Step Costs:** A → B (2), A → C (5), B → D (4), B → E (7), C → F (2), D → G (3), F → G (6), E → H (5), G → H (2).
- **Heuristic Values h(n):** A = 10, B = 8, C = 7, D = 6, E = 4, F = 5, G = 2, H = 0.

a. Expand the nodes and **explain** the order of their expansion using Greedy Best-First Search and A* Search.

b. Interpret the path returned by Greedy Best-First Search and A* Search.

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Question No.4 [CLO-2]

[Marks: 12]

Discuss why local search algorithms are useful for optimization problems, and describe the different types of local search techniques. Then, using a Genetic Algorithm work with the following initial population of chromosomes for an optimization task:

Initial Population: 28, 43, 17, 61

Question No.5 [CLO-2]

[Marks: 12]

Interpret Machine Learning (ML) and its main types, and **discuss** how each type contributes differently to AI problem-solving.

Then, consider the following scenario:

A hospital deploys an ML model to flag potentially dangerous drug-drug interactions when a physician writes a new prescription. The model reviews newly prescribed drug pairs and raises an alert if it predicts a harmful interaction. These dangerous interactions are relatively rare.

There is a well-established clinical verification layer: pharmacists manually review flagged prescriptions, and interaction-checking rule software is already in place.

However, raising too many false alarms (i.e., false positives) is very costly, as it delays patient care and causes alert fatigue in doctors and pharmacists.

a. Illustrate how each type of ML can be applied within this scenario.

b. In this scenario, which performance metric is more suitable? **Infer** two reasons to support your answer.

c. Explain why achieving high accuracy on training data does not guarantee strong real-world performance.

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Max Marks : 60

Fundamentals of Data Science - CT-368

Note: Phone/ Smart Watch are not allowed in examination Room.

Q#1a **Discuss** the concept of Data Science? Discuss the difference between the skill set of a Data Scientist, Data Engineer and a Data Analyst. Mention tools required as well. CLO-1 (6 mark)

Q#1b **Explain** the difference between Binary Encoding and One-Hot Encoding. Given the categorical feature Color = ['Red', 'Green', 'Blue'], show how it can be encoded using both methods. Write Python code to implement both encodings using pandas or scikit-learn. CLO-1 (6 marks)

Q#2a **Discuss** Data transformation techniques. You are given the following dataset representing the ages and incomes of 5 people: CLO-1 (6 marks)

Person	Age (years)	Income (USD)
1	25	50000
2	30	60000
3	45	70000

- Scale the Age and Income values using normalization to a range of [0,1]
- Standardize the Age and Income values using z score techniques.

Q#2b **Discus** how random forests solve disadvantages of decision trees. Calculate the missing values using proximity matrix and random forest algorithm. Interpret the result. CLO-1 (6 marks)

Purchase Frequency	Payment Method	Loyalty Tier	Monthly Spending ($)	Will Purchase Again
High	Credit Card	Gold	420	Yes
Low	Cash	Silver	120	No
Medium	Credit Card	Gold	350	Yes
High	?	?	?	Yes

	C1	C2	C3	C4
C1	1.0	0.2	0.7	0.8
C2	0.2	1.0	0.3	0.1
C3	0.7	0.3	1.0	0.9
C4	0.8	0.1	0.9	1.0

Q#3 **Discuss** whether SVM is robust to outliers or not. How do hard margin SVM and soft margin SVM handle outliers differently? Give one example of three commonly used kernel. CLO-1 (6 marks)

Q#4a **Compare** type 1 and type 2 error. A bank develops a credit risk model to classify loan applicants into four risk categories: (Low Risk, Medium Risk, High Risk, Very High Risk). After testing the model on 1000 applicants, the confusion matrix is as follows: CLO-2 (6 marks)

Actual \ Predicted	Low Risk	Medium Risk	High Risk	Very High Risk
Low Risk	180	10	5	5
Medium Risk	20	140	15	5
High Risk	5	15	130	10
Very High Risk	10	5	15	160

1. Calculate Accuracy, Precision and Recall for the "Very High Risk" class.
2. The bank wants to avoid approving loans for applicants who are likely to default (Very High Risk). Which metric should the bank prioritize? Explain why.

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Q#4b **Analyze** bias and variance. Given the weather data, with counts and probabilities. CLO-2 (6 marks)

weather			temperature			humidity			windy			play	
	yes	no		yes	no		yes	no		yes	no	yes	no
sunny	2	3	hot	2	2	high	3	4	FALSE	6	2	9	5
cloudy	4	0	mild	4	2	normal	6	1	TRUE	3	3		
rainy	3	2	cold	3	1								

Predict the output using Naive bias algorithm. Discuss the difference between Probability and likelihood

weather	temperature	humidity	windy	play
sunny	Cold	high	true	?

Q#5a **Analyze** Principal Component Analysis (PCA) with an example. List the main steps involved in performing PCA on a dataset. Explain one advantages and one disadvantage of PCA. CLO-2 (6 marks)

Q#5b **Compare** how is datafication different from digitization and explain two major ethical concerns that arise in Data Science related to privacy and security. Provide examples. CLO-2 (6 marks)

Q#6 **Compare** the following CLO-2 (6 marks)

- K-mean and K-medoid clustering
- Linear regression and Logistic Regression

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Time : 3 Hours | Batch 2023 | Dated : 10-DEC-25

Max Marks : 60

Introduction to Tools and Techniques for Data Science - CT-265

Note: Solve in chronological order.

Question 1 [CLO-1]

a) **Explain** the significance of data in modern applications and the insights it can give in any **three** domains (e.g., healthcare, finance, transportation, e-commerce). (6 marks)

b) **Illustrate** with one detailed example how raw data is converted into a useful insight for decision-making in any one of these domains. (4 marks)

Question 2 [CLO-2]

Analyze the dataset of the following students:

Student	Marks (%)	Study Hours per Week	City
A	45	5	Karachi
B	82	10	Lahore
C	60	7	Karachi
D	92	12	Islamabad
E	35	3	Karachi

a) Inspect the dataset and identify the types of data for each column. (3 marks)

b) Analyze which pre-processing steps would be appropriate if this dataset were to be used in a machine learning model. Justify each chosen step briefly. (4 marks)

c) Examine and suggest two suitable visualizations for this data and state what insight each visualization could reveal. (3 marks)

Question 3 [CLO-2]

Analyze the dataset of the following students:

Student	Study Hours per Week (X)	Marks (%) (Y)
A	2	40
B	4	50
C	6	60
D	8	70
E	10	80

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a) Using min–max normalization, transform the Marks column to the range [0,1] (4 marks)
b) Standardize the Marks using the z-score formula. (3 marks)
c) Using the following bins, categorize each student into a performance group based on Marks. Write the bin for each student. (2 marks)
   a. Low: 0–50%
   b. Medium: 51–70
   c. High: >70%
d) Calculate the Pearson correlation coefficient (r) between Study Hours (X) and Marks (Y). (4 marks)
e) Based on the value of r, interpret and analyze the relationship between Study Hours and Marks. (2 marks)

Question 4 [CLO-2]

Analyze the following **HTML snippet** from an online store showing customer reviews for a mobile phone:

```
<div class="review">
 <p class="text">Delivery was fast and the phone quality is excellent.</p>
 <span class="rating">5</span>
</div>

<div class="review">
 <p class="text">Very bad packaging, the box was damaged and late delivery.</p>
 <span class="rating">1</span>
</div>

<div class="review">
 <p class="text">Good value for money, battery life is great.</p>
 <span class="rating">4</span>
</div>

<div class="review">
 <p class="text">Terrible service, slow delivery and poor response from support.</p>
 <span class="rating">1</span>
</div>
```

Assume:

- Reviews with rating ≥ 4 are positive.
- Reviews with rating ≤ 2 are negative.
- Use the following stop word list in pre-processing:
  ["the","is","am","are","a","an","and","but","very","was","too","of","for","to","from"]

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a) Write Python code to: (5 marks)
   a. Parse the HTML.
   b. Find all reviews blocks.
   c. For each block, extract:
      i. The review text inside
      ii. The rating inside (as an integer)

b) Using the extracted reviews: (8 marks)
   a. Form two corpora after pre-processing i.e. Positive corpus and Negative corpus
   b. Write the final token lists for positive corpus and negative corpus.
   c. Create a vocabulary table that contains each unique word appearing in *any* corpus and its frequency in positive and frequency in negative reviews.

c) Classifying a new review using **Logistic Regression**. Review: "**Fast delivery and great battery life but poor packaging**." (7 marks)

The model parameters are:
$\theta = [\theta_0, \theta_1, \theta_2] = [-1.5, 0.6, -0.7]$

d) An e-commerce site uses product codes of the form: (5 marks)

- Start with the exact letter's PRD.
- Followed by a hyphen -
- Followed by 3 to 5 digits (0–9).
- No other characters before or after.

   a. Write a regular expression that matches exactly such valid product codes.
   b. Using your regex, state which of the strings in the list would match and which would not match. [PRD-123, PRD-98765, PROD-1234, XPRD-12345, PRD-12AB]

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Max Marks : 60

Platform & Architecture for Data Science - CT-370

- Note: Solve in chronological order.

Question 1 [CLO-1]

Define in detail that four types of analytics and also draw its hierarchy pyramid? (12 Marks)

Question 2 [CLO-1]

Differentiate in detail between Traditional data and Big Data, also **Explain** the 5 vs of big data? (12 Marks)

Question 3 [CLO-2]

Explain Platforms, Tools and Environments also **Compare** with the help of table? (12 Marks)

Question 4 [CLO-2]

Explain Data Science Lifecycle and also **Draw** its flow diagram? (12 marks)

Question 5 [CLO-3]

a. **Construct** the end-to-end data flow in Uber's data pipeline along with its roles, and show how these stages ensure real-time decision-making also draw its flow diagram. (6 Marks)

b. **Illustrate** the core components and real-world instances of MODERN DATA STACK? (6 Marks)

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